

ABHISHEK PRAKASH

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EDUCATION

Northeastern University, Boston, MA

January 2024 - May 2026

Master of Science in Computer Software Engineering

Coursework: Application Engineering & Development, Cloud Computing, Algorithms, Applied Machine Learning

VIT University, Vellore, India

August 2018 - May 2022

Bachelor of Technology in Mechanical Engineering

Coursework: Big Data, Data Structure and Algorithm, Operating System, Computer Network, Object Oriented Design

WORK EXPERIENCE

Software Development Co-op | Dassault Systèmes SolidWorks, Waltham, MA

January 2025 - September 2025

- Modernized 4+ legacy Perl/C# applications to TypeScript microservices on Kubernetes, enabling faster feature releases and reducing deployment disruptions, improving experience for 10K+ daily CAD users
- Designed centralized REST API service with Swagger documentation following DRY principles, eliminating code duplication across applications and improving API latency by 85% (3s to 450ms)
- Architected centralized logging infrastructure using Fluent Bit and Elasticsearch, achieving 100% log visibility across all microservices and reducing debugging time by 75% after identifying critical observability gaps
- Built internal NPM package with automated CI/CD pipeline publishing reusable utilities and API clients, reducing code duplication by 70% across microservices and standardizing error handling patterns

Software Developer | Deloitte, Hyderabad, India

May 2022 - December 2023

- Built full-stack insurance platform with Python/FastAPI microservices and React frontend processing 1M+ transactions for policy management with 99.9% uptime, supporting 20K+ insurance agents across 30 offices
- Implemented event-driven architecture with RabbitMQ message broker connecting 8 microservices, enabling real-time data synchronization between policy, billing, and customer systems with guaranteed message delivery
- Optimized API performance by implementing Redis caching layer and Celery task queues for batch processing, reducing average response times by 60% while handling 5K+ concurrent users in production
- Established CI/CD pipelines with Jenkins and Docker containerization, automating deployments across environments with rollback capabilities, cutting release cycles from bi-weekly to daily

Software Developer Intern | Zomato, Bangalore, India

May 2020 - September 2020

- Enhanced React frontend performance working with Java Spring Boot backend, reducing search query response time by 30% through optimized component rendering and API payload compression serving 50K+ daily users
- Revamped restaurant search functionality using Elasticsearch with fuzzy matching algorithms and synonym mapping, improving search success rate from 65% to 92% for misspelled queries
- Built real-time order tracking with WebSocket connections and Redis pub/sub messaging, enabling live status updates for delivery drivers and customers across 10K+ concurrent active orders

ACADEMIC PROJECTS

E-Commerce Search Platform | Node.js, Elasticsearch, React, PostgreSQL, Redis **August 2024 - November 2024**

- Built product search engine using Node.js with Elasticsearch, implemented real-time inventory sync across 3 warehouses preventing overselling, processed 10K+ daily queries with autocomplete and faceted filtering
- Developed React storefront with product recommendations based on browsing patterns cached in Redis, integrated PostgreSQL for order management, deployed with Docker

Microservices Banking Platform | AWS, Terraform, Java, PostgreSQL

October 2024 - December 2024

- Automated AWS cloud infrastructure with Terraform provisioning ECS clusters and RDS PostgreSQL, configured GitHub Actions CI/CD pipeline validating infrastructure changes before production deployment
- Built Java Spring Boot microservices for account management and transactions deployed on ECS with auto-scaling, integrated CloudWatch monitoring for API performance with automated SNS alerts

Real-Time Fraud Detection | Python, FastAPI, PostgreSQL, Redis, Kafka

October 2025 - December 2025

- Implemented transaction fraud detection pipeline using Kafka streaming payment events to Python consumers, analyzing velocity patterns and flagging suspicious activity with Redis caching for user history lookups
- Trained scikit-learn model on historical transaction data with SMOTE handling class imbalance, achieving 92% precision identifying fraudulent transactions with automated alerts via FastAPI endpoints

TECHNICAL SKILLS

Languages & Frameworks: Java, Spring Boot, Python, FastAPI, TypeScript, JavaScript, React, Node.js

Cloud & DevOps: AWS (ECS, RDS, CloudWatch), Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD

Data & AI: PostgreSQL, Redis, Kafka, RabbitMQ, Celery, GraphQL, Apollo Server, scikit-learn, Pandas